

04. FORMULAE; CONVERSIONS; VALUES, UNITS. (120) (2023 – 2024).

01. FORCE.

- | | |
|---------------------------------|-----------------------------|
| 1. F | = ma |
| 2. F | = mg |
| 3. Weight | = mg |
| 4. Moment of force | = $F \times \perp$ distance |
| 5. Moment of couple | = $F \times$ diameter |
| 6. Sum of anticlockwise moments | = Sum of clockwise moments |

02. WORK, ENERGY & POWER.

- | | |
|-------------------|--|
| 7. F | = ma |
| 8. F | = Weight = mg |
| 9. W | = $F \times S$ (horizontal motion) |
| 10.W | = $F \times S \cos \theta$ (angular motion) |
| 11.W | = mgh (Vertical) |
| 12.Energy | = Work |
| 13.P | = $\frac{W}{t}$ |
| 14.P | = $F \times v$ (horizontal motion) |
| 15.P | = $F \times v \cos \theta$ (angular motion) |
| 16.Efficiency | = $\frac{\text{power output}}{\text{power input}}$ |
| 17.V | = $u \pm at$ |
| 18.S | = $\frac{1}{2}(u + v) t = ut \pm \frac{1}{2}at^2$ |
| 19.v ² | = $u^2 \pm 2aS$ |
| 20.F | = ma |

21.W	= U	= K
22.W		= FS
23.U		= mgh
24.Gain in U		= mg (h ₂ – h ₁)
25.Loss in U		= mg (h ₁ – h ₂)
26.K		= $\frac{1}{2} mv^2$
27.K		= $\frac{p^2}{2m}$
28.Δ K		= $\frac{1}{2} m (v_2 - u_2)$
29.p		= m v
30.p		= $\sqrt{2mK}$
31.W		= mgh
32.U		= mgh
33.Δ in U		= mg (h ₂ – h ₁)
34.K.E.		= $\frac{1}{2} mv^2$
35.Δ in K.E.		= $\frac{1}{2} (mv^2 - mu^2)$
36.Impulse		= m (v – u)

03. MACHINES.

37.Mechanical advantage (M.A.)	= $\frac{\text{Load (L)}}{\text{Effort (E)}}$ = $\frac{\text{Effort arm}}{\text{Load arm}}$
38.Velocity ratio (V.R.)	= $\frac{\text{Displacement of the effort}}{\text{Displacement of the load}}$ = $\frac{d_E}{d_L}$
39.Work input	= E × d _E
40.Work output	= L × d _L

41. Efficiency (η)	$= \frac{\text{Work output}}{\text{Work input}}$ $= \frac{L}{E} \times \frac{d_L}{d_E}$ $= \frac{\text{Power output}}{\text{Power input}}$ $= \frac{MA}{VR}$
42. Efficiency (η)%	$= \frac{M.A.}{V.R.} \times 100$
43. Load $\times$ Load arm	$= \text{Effort} \times \text{Effort arm}$
44. Load $\times$ Load arm	$= \text{Effort} \times \text{Effort arm}$
45. Mechanical advantage (M.A.)	$= \frac{\text{Load (L)}}{\text{Effort (E)}} = \frac{\text{Effort arm}}{\text{Load arm}}$
46. Velocity ratio (V.R.)	$= \frac{\text{Displacement of the effort}}{\text{Displacement of the load}} = \frac{d_E}{d_L}$
47. Work input	$= E \times d_E$
48. Work output	$= L \times d_L$
49. Efficiency (η)	$= \frac{\text{Work output}}{\text{Work input}} = \frac{L}{E} \times \frac{d_L}{d_E} = \frac{\text{Power output}}{\text{Power input}} = \frac{MA}{VR}$
50. Resistance due to movable parts or friction of the pulley (X) = E (VR – MA)	
	OR
	$= VR - MA$
51. VR	$= \frac{2R}{R - r}$

04. REFRACTION OF LIGHT AT PLANE SURFACES.

52. ${}_a\mu_g$	$= \frac{\text{speed of light in vacuum (c)}}{\text{speed of light in that medium (v)}}$
53. ${}_a\mu_g$	$= \frac{\sin \text{ of angle of incidence (sin } i \text{)}}{\sin \text{ of angle of refraction (sin } r \text{)}}$
54. ${}_1\mu_2$	$= \frac{\text{speed of light in medium 1}}{\text{speed of light in medium 2}}$
55. ${}_1\mu_2 \times {}_2\mu_1$	$= 1$
56. ${}_1\mu_2$	$= \frac{\lambda_2}{\lambda_1}$

$$\begin{aligned}
 57. \delta_{\text{Min}} &= 2i - A \\
 58. \delta &= (i_1 + i_2) - (r_1 + r_2) \\
 59. r_1 + r_2 &= A \\
 60. i_1 + i_2 &= A + \delta \\
 61. \text{Refractive index} &= \frac{\text{Real thickness}}{\text{Apparent thickness}} = \frac{RD}{AD} \\
 62. \text{Shift} &= RD \left(1 - \frac{1}{\mu}\right) \\
 63. \text{Shift} &= RD - AD
 \end{aligned}$$

05. REFRACTION THROUGH A LENS.

$$\begin{aligned}
 64. \text{Power of lens (in D)} &= \frac{1}{\text{focal length (in meters)}} \\
 65. \frac{1}{f} &= \frac{1}{v} - \frac{1}{u} \\
 66. \text{Magnification, } m &= \frac{\text{height of image (I)}}{\text{height of the object (O)}} = \frac{v}{u} \\
 67. &
 \end{aligned}$$

Concave	Convex	
always virtual	Virtual	Real
-u	-u	-u
-v	-v	+v
-f	+f	+f
+m	+m	-m

06. SPECTRUM.

$$\begin{aligned}
 68. c &= f \lambda \text{ (for light ray)} \\
 69. v &= f \lambda' \text{ (for any other ray)} \\
 70. c &= f \lambda \\
 71. v &= f \lambda'
 \end{aligned}$$

07. SOUND.

72.t

$$= \frac{\text{Total distance travelled}}{\text{Speed of sound}}$$

$$= \frac{2d}{v}$$

73.f

$$= \frac{1}{T}$$

74.T

$$= \frac{1}{f}$$

75.d

$$= \frac{vt}{2}$$

$$= \frac{v}{2f}$$

76.v

$$= \frac{2d}{t}$$

$$= 2df$$

77.v

$$= f \lambda$$

78. $l_1 f_1$

$$= l_2 f_2$$

79. $f_1 \lambda_1$

$$= f_2 \lambda_2$$

08. CURRENT ELECTRICITY.

80.V

$$= I R$$

81.I

$$= \frac{V}{R}$$

82.R

$$= \frac{V}{I}$$

83.R

$$= \rho \frac{l}{a}$$

$$= \rho \frac{l}{\pi r^2}$$

84. ρ

$$= \frac{R a}{l}$$

$$= \frac{R \pi r^2}{l}$$

85.I

$$= \frac{Q}{t}$$

$$= \frac{ne}{t}$$

86.W

$$= QV$$

87.V

$$= I R \text{ (external)}$$

88.V

$$= I r \text{ (internal)}$$

89.V

$$= IR$$

90. Resistivity ρ	$= \frac{Ra}{l}$
91. Total resistance of the circuit	$= R + r$
92. Current drawn from the cell I	$= \frac{\varepsilon}{R+r}$
93. Terminal voltage of the cell V	$= IR$ $= \frac{\varepsilon R}{R+r}$
94. Voltage drop inside the cell	$v = Ir$ $= \frac{\varepsilon r}{R+r}$
95. Internal resistance of the cell, r	$= \left(\frac{\varepsilon}{V} - 1 \right) R$ $= \frac{v}{I}$
96. E.m.f of cell, ε	$= I (R + r)$
97. Equivalent resistance in series R_s	$= R_1 + R_2 + R_3 + \dots$
98. Equivalent resistance in parallel $\frac{1}{R_p}$	$= \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \dots$
99. I	$= \frac{Q}{t}$ $= \frac{ne}{t}$
100. Charge of 1 electron	$= -1.6 \times 10^{-19} \text{ C}$
101. 1 A current carries 6.25×10^{18} elec.	
102. W	$= QV = V I t = I^2 R t = \frac{V^2 T}{R} = P \times t$
103. P	$= \frac{W}{t} = \frac{QV}{t} = V I = I^2 R = \frac{V^2}{R}$
104. Q	$= It$
105. H	$= 0.24 I^2 R t$

09. HOUSEHOLD CIRCUITS. – NO NUMERICALS

10. ELECTROMAGNETISM.

106. Turns ratio $= \frac{N_s}{N_p}$
107. $\frac{N_s}{N_p}$ $= \frac{E_s}{E_p}$
108. E_s, I_s $= E_p, I_p$
109. ΔB $=$ Change in magnetic flux
110. emf $= N \times \frac{\Delta B}{t}$

11. CALORIMETRY.

111. Q $= mc \Delta T$
112. c $= \frac{Q}{m \Delta T}$
113. c $= \frac{P \times t}{m \Delta T}$
114. C' $= c \times m$
115. C' $= \frac{Q}{\Delta T}$
116. Q $= mL$
117. L $= \frac{\text{Heat absorbed/liberated during the change of phase}}{\text{mass}}$
- $= \frac{Q}{m}$
- $= \frac{Pt}{m}$

12. RADIOACTIVITY.

118. 1 a.m.u. $= 931 \text{ MeV}$
119. E $= mc^2$
120. E $= \text{loss of mass in a.m.u} \times 931 \text{ MeV}$

CONVERSIONS:

1. 1 calorie	= 4.2 joules
2. 1 kilocalorie = 1000 calories	= 4200 J
3. 1 cal K ⁻¹	= 4.2 JK ⁻¹
4. 1 kilocalorie °C ⁻¹	= 1000 calorie °C ⁻¹
5. 1 cal g ⁻¹ °C ⁻¹	= 4.2 × 10 ³ JKg ⁻¹ K ⁻¹
6. 1 kilo-cal kg ⁻¹	= 1 cal g ⁻¹
7. 336000 J Kg ⁻¹	= 80 cal g ⁻¹
8. 1 Kgf	= 9.8 N,
9. 1gf	= 980 dynes.
10. 1g/cm ³	= 1000 kg/m ³
11. 1 Newton	= 10 ⁵ dynes
12. 1 bar	= 10 ⁵ Pascal
13. 1Km / hour	= 5/18 m/s
14. 1 liter	= 1000 cm ³
15. 1 m ³	= 1000 liters
16. 1 Kg	= 1000 g
17. 1 atmospheric pressure	= 76 cm mercury column
18. Density of water	= 1 g /cm ³ (C.G.S), 1000 Kg / m ³ (S.I.)
19. Density	= R.D (C.G.S), density = 1000 × R.D.
20. 1 joule	= 10 ⁷ erg
21. 1 Kw	= 1000 w
22. 1 Mw	= 10 ⁶ W
23. 1 k J	= 10 ³ J
24. 1 joule	= 10 ⁷ erg
25. Watt	= 1 J s ⁻¹
26. 1 W	= 10 ⁷ erg s ⁻¹

27. 1 HP	= 746 W	= 0.764 kW
28. 1 joule		= 0.24 calorie
29. 1 eV		= 1.6×10^{-19} J
30. 1 Wh	= 3600 J	= 3.6 kJ
31. 1 KWh	= 3.6×10^6 J	= 3.6 MJ

VALUES:

1. Specific heat capacity of water = $4.2 \text{ Jg}^{-1} \text{ K}^{-1}$
2. Specific latent heat of steam = 2268 Jg^{-1}
3. Specific heat capacity of fusion of ice = 336 Jg^{-1}
4. Specific heat capacity of ice = $2.1 \text{ Jg}^{-1} \text{ K}^{-1}$

UNITS:

Physical quantities	C.G.S.	S.I.
Area	cm^2	m^2
Volume	cm^3	m^3
Velocity/speed	cm/s	ms^{-1}
Acceleration	cm/s^2	ms^{-2}
Density	g/cm^3	Kg m^{-3}
Momentum	g.cm/s	Kg ms^{-1}
Force / weight / thrust / upthrust / load / effort	Dyne	newton or Kg ms^{-2}
Work output / Work input / energy	Erg	Joule or $\text{Kg m}^2\text{s}^{-2}$
Power	erg/s	Joule (Watt) or $\text{Kg m}^2\text{s}^{-3}$
Pressure	dyne/cm^2	newton/m^2 (pascal)
Electric current	-----	Ampere
Temperature	-----	Kelvin
Mass	Gram	Kilogram
Length	cm	M
Amount of substance	-----	Mole
Intensity of light	-----	Candela

Note: gravitational units of weight / upthrust: gram force, kilogram force.

SCALAR AND VECTOR QUANTITIES:

No direction	Has direction
(1) Speed	(a) Velocity
(2) Mass	(b) Momentum
(3) Work	(c) Displacement
(4) Distance	(d) Acceleration
(5) Frictional Forces	(e) Force (thrust): downward
(6) Power	(f) Weight: downward
(7) Energy	(g) Gravitational force: downward
(8) Pressure	(h) Upthrust: upward
(9) Frequency	(i) Tension: upward

@icse10channel